

Graded potentials

The written notes for the graded potentials walkthrough. Week 4 competencies 12 and 14.

To print: these notes as a two-column PDF.

How to use these notes. Work through the walkthrough first, writing each prediction on your worksheet before you press Show me. Then read these notes to fill in anything you missed, and use them to review. They are the same facts in a form you can study from, not a copy of the slides.

Two kinds of electrical signal

	Graded potential	Action potential
Where	Input zone: dendrites and cell body	Starts at the trigger zone, travels the axon
Size	Varies with the stimulus	Always the same size (all or none)
Distance	Short; fades as it spreads	Long; does not fade
Threshold	None	Fires only if the trigger zone reaches threshold
Channels	Chemically or mechanically gated	Voltage-gated sodium and potassium

How a graded potential is made

1. A stimulus opens gated channels: a neurotransmitter opens ligand-gated channels at a synapse, or stretch opens mechanically gated channels in a sensory ending.
 2. Ions cross, and the membrane moves toward the equilibrium potential of the ion whose permeability went up.
 3. Sodium in: the inside becomes less negative, a depolarizing graded potential.
 4. Potassium out: the inside becomes more negative, a hyperpolarizing graded potential, which moves the cell away from firing.
 5. The channels close and the membrane returns to rest. A graded potential lasts from a few milliseconds to tens of milliseconds.
- A stronger stimulus opens more channels or holds them open longer, so more ions cross and the change is larger. That is what graded means.
 - Example sizes recorded at one spot: weak stimulus 3 mV, medium 7 mV, strong 12 mV; a potassium-opening stimulus 8 mV below rest.

Why a graded potential fades as it spreads

- Local current flow: the positive charge that enters spreads along the inside of the dendrite in both directions.
- Current leak: some of that charge escapes through the open leak channels along the membrane, like water from a leaky hose.
- Cytoplasmic resistance: the cytoplasm resists the flow of charge, the way a thin wire resists current more than a thick one.

- Together they make the change smaller at every point farther from the channels that opened. Example: 20 mV at point 1 next to the channels, then 12, 7 and 4 mV at points farther along.
- A wider dendrite has lower cytoplasmic resistance, so a graded potential spreads farther. Current leak does not change, because it depends on the leak channels in the membrane.

Local current flow

Spread of charge through the cytoplasm away from where ions entered.

Current leak

Loss of charge across the membrane through open leak channels.

Cytoplasmic resistance

The cytoplasm's resistance to the flow of charge; lower in a wider process.

What reaches the trigger zone

- The trigger zone fires only if the membrane there reaches threshold, about -55 mV. From rest at -70 mV that takes about 15 mV of depolarization.
- What counts is the size of the graded potential when it arrives, not where it started.
- Graded potentials that overlap in time add together: summation. Depolarizing ones add; hyperpolarizing ones subtract.
- Real graded potentials that overlap add to a little less than their plain sum. The reasoning is the same.

Inputs reaching the trigger zone	Arithmetic	Voltage	Result
One excitatory, 8 mV	$-70 + 8$	-62 mV	Subthreshold, no action potential
Two excitatory together, 8 mV each	$-70 + 16$	-54 mV	Suprathreshold, fires
Two excitatory plus one inhibitory of 5 mV	$-70 + 8 + 8 - 5$	-59 mV	Subthreshold, no action potential

Threshold

The voltage at the trigger zone, about -55 mV, at which an action potential fires.

Subthreshold

A graded potential too small to bring the trigger zone to threshold.

Suprathreshold

A graded potential large enough to bring the trigger zone to threshold.

Summation

The adding together of graded potentials that overlap in time.

Patient: the pufferfish meal

- Tetrodotoxin plugs voltage-gated sodium channels from the outside. It does not block chemically or mechanically gated channels.
- His resting potential is normal (-70 mV). Weak stimuli still make small graded potentials, about 5 mV, that fade.
- A strong stimulus makes about 20 mV, which goes past threshold, but no action potential appears, because the channels that make the action potential are plugged.
- His neurons can still take in signals but cannot send them down an axon. The action potential walkthrough finishes this case.